

Kian Kyars

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Experience

Judgment Labs

March 2026 - August 2026

Research Engineer

- Created a dynamic KNN+MST algorithm for clustering agent traces
- Designed a self-improvement harness for software-engineering agents using Judgment's tracing platform
- Implemented distributed PII-redaction on traces and a token-efficient MCP server

Noemon

October 2025 - January 2026

Machine Learning Engineer

- Developed an ARC-AGI-2 refinement solver with a ReAct approach

fr8 Cohort 1.0

June 2025 - August 2025

- Landed [35 merged PRs](#) to [Factorio Learning Environment](#) across agent evaluation, Gym infrastructure, and testing

Data-Driven Network and Cyber Security (DANS) Lab

September 2024 - June 2025

Undergraduate Research Assistant

- Fine-tuned Qwen2.5, Llama 3, and DeepSeek-R1 7-8B models with unsloth for provenance graph construction
- Performed adaptive anomaly detection locally on system logs with equivalent performance to GPT-4o

MANGA NLP Lab

January 2025 - April 2025

Undergraduate Research Assistant

- Reimplemented the FLORA optimizer, improving 3B/7B LLM training speed up to 25% versus LoRA with AdamW

Hochschule für Technik und Wirtschaft Dresden

May 2023 - August 2023

Summer Research Intern

- DAAD RISE Intern; trained a 7-DOF Franka Emika arm in Nvidia Isaac Sim for block manipulation using TRPO and PPO

Projects

CHIPS: From Design to Data Center

July 2026

- Published a ~3-hour course on the semiconductor supply chain, from chip design to data-center deployment

Reinforcement Learning from Verifiable Rewards (textbook)

April 2026

- Authored a 10-chapter reference on verifier design, reward hacking, training signals, and agentic RLVR

OpenClaw Course

February 2026

- Tutorial on setting up an autonomous agent & messaging gateway with OpenClaw; 465k+ views

Pipeline Parallelism from Scratch

January 2026

- Implemented a PyTorch pipeline-parallel trainer from scratch ([micropp](#)): model partitioning, send/recv primitives, and naive, GPipe, and 1F1B schedules; 3.5-hour freeCodeCamp course (38k+ views)

Data Parallelism from Scratch

February 2026

- Implemented distributed data parallel training from scratch ([microddp](#)): all-reduce, gradient bucketing, and overlapped communication; 1.75-hour freeCodeCamp course (19k+ views)

nanochat from Scratch

October 2025

- Re-implemented and trained nanochat on an 8XH100 node with DeepSpeed ZeRO-2

Hugging Face Ultra Scale Playbook Series

June 2025

- Created a [16-part video series](#) covering the Hugging Face Ultra-Scale Playbook; recognized by Hugging Face CSO [Thomas Wolf](#), who signed the book

Founder - UofACarpool

March 2024 - April 2025

- Wrote algorithm and website for matching university commuters to carpool together; featured in the [Gateway](#)

Education

BSc Computer Science, Honours - University of Alberta

C/O 2025 *Grade: 3.9/4.0*

Interests

- Polyglot (English, French, Farsi, German) and ultrarunner (longest distance: [108.67 km](#))